Project Synopsis: Generative AI System for Context-Aware Complaint Prioritization

---

0. Cover

- **Project title:** Generative AI System for Context-Aware Complaint Prioritization
- **Team name & ID:** $(LOGIC)^3$ - T34
- **Institute / Course:** GLA University, Department of Computer Engineering and Applications, Mini Project
- **Version:** v1.0
- **Date:** 06 February 2026

Revision history

Version	Date	Author	Change
v0.1	05 Feb 2026	Gaurav Chaudhary	Initial draft
v1.0	06 Feb 2026	Team $(LOGIC)^3$	Finalized Synopsis

---

1. Overview

- **Problem statement:** Organizations receive a massive volume of customer complaints through emails, web portals, chats, call transcripts, and social platforms. Existing complaint handling systems rely heavily on keyword matching and basic sentiment analysis, which fail to capture the true urgency, business impact, and systemic risk of complaints.
- **Goal:** To design and develop a Generative AI-driven, context-aware complaint prioritization system that intelligently analyzes complaint text, infers urgency and impact, detects systemic risks, assigns priority scores, and provides explainable justifications for escalation decisions.

- **Non-goals:** Automated customer replies or chatbots, Real-time voice call processing (v1), Fully autonomous resolution handling, Legal decision-making.
- **Value proposition:** Contextual reasoning beyond sentiment, Early detection of systemic failures, Transparent and explainable prioritization, Reduced resolution time and operational cost, Scalable AI-assisted decision support.

---

2. Scope and Control

2.1 In-scope

- Complaint ingestion and preprocessing, NLP-based semantic understanding, LLM-driven contextual reasoning, Urgency, impact, and risk inference, Priority scoring engine, Explainable escalation logic, Feedback loop from resolution outcomes, Analytics dashboard, REST APIs using FastAPI.

2.2 Out-of-scope

- Payment dispute resolution, Automated legal compliance enforcement, Multimodal inputs (images/audio) in v1, Fully automated decision enforcement.

2.3 Assumptions

- Complaints are text-based
- Historical complaint data is available
- Human review exists for escalated cases
- English language complaints (v1)

2.4 Constraints

- Limited development timeline
- Restricted compute resources
- API rate limits for LLM usage

- Academic project boundaries

2.5 Dependencies

- Pre-trained LLM (GPT / Gemini / LLaMA), Vector database for embeddings, Database for complaint storage, Visualization framework for dashboards

2.6 Acceptance criteria and sign-off

- GIVEN a complaint with financial risk WHEN processed THEN priority score ≥ threshold
- GIVEN multiple similar complaints WHEN clustered THEN systemic risk is flagged
- **Sign-off:** Project demonstration, documentation, and test report approved by faculty mentor.

---

3. Stakeholders and RACI

Activity	**Responsible (R)**	**Accountable (A)**	**Consulted (C)**	**Informed (I)**
Requirement Analysis	Gaurav	Gaurav	Mentor	Team
System Design	Team	Gaurav	Mentor	Team
Development	Team	Gaurav	Mentor	Dept
Testing	Ichcha	Gaurav	Mentor	Team
Documentation	Balkrishna	Gaurav	Mentor	Dept

---

4. Team and Roles

Member	Role	Responsibilities
Gaurav Chaudhary	Team Lead & Architect	System design, APIs, LLM logic
Ichcha Mehrishi	NLP & Analysis	Feature extraction, evaluation
Balkrishna Goswami	Backend & Dashboard	FastAPI, analytics, data flow

---

5. Week-wise Plan and Assignments

Week	Milestones	Deliverables
1	Requirement analysis	SRS
2	Architecture design	Diagrams
3	NLP preprocessing	Feature extractor
4	LLM integration	Reasoning engine
5	Priority scoring	Scoring module
6	Explainability	Explanation logic
7	Dashboard & APIs	UI + API

Week	Milestones	Deliverables
8	Testing & finalization	Report & demo

---

6. Users and UX

6.1 Personas

- **Support Agent:** Needs quick prioritization
- **Operations Manager:** Needs systemic insights
- **Admin:** Needs transparency and auditability

6.2 Top user journeys

Complaint → AI Analysis → Priority Score → Escalation → Resolution → Feedback

6.3 User stories

- As a manager, I want to know *why* a complaint was escalated
- As an agent, I want critical complaints first

6.4 Accessibility & localization

- Web-based dashboard, Simple visual cues, English language support

---

7. Market and Competitors

7.1 Competitor table

System	Approach	Limitation
Zendesk	Rules + sentiment	No deep reasoning
Freshdesk	Keyword-based	Poor explainability
Our System	Context-aware AI	Intelligent escalation

7.2 Positioning

- **A decision-support AI system, not a chatbot.**

---

8. Objectives and Success Metrics

Objective	KPI
Accurate prioritization	≥ 85% precision
Systemic risk detection	Early incident flag
Explainability	Human-understandable
Resolution efficiency	Reduced SLA breach

---

9. Key Features

Feature	Description
Urgency inference	Time-critical reasoning
Impact analysis	Business & user impact
Systemic risk	Pattern detection
Explainability	Transparent logic
Feedback loop	Learning system

10. Architecture

10.1 High-level

- Frontend Dashboard
- FastAPI Backend
- NLP Feature Extractor
- LLM Reasoning Engine
- Priority Scoring Engine
- Database & Analytics Layer

10.2 API spec snapshot

Endpoint	Purpose
/complaints	Ingest complaint
/analyse	AI reasoning
/score	Priority calculation
/dashboard	Analytics

10.3 Config and secrets

- Environment variables, API key security, Access control.

---

11. Data Design

11.1 Data dictionary

The system uses a structured data model to represent complaints, AI analysis outputs, prioritization scores, and feedback signals.

Entities and Attributes: -

Complaint Entity

Field	Type	Description
complaint_id	UUID	Unique identifier
complaint_text	Text	Raw complaint content
source	String	Email, portal, chat, etc.
created_at	Timestamp	Complaint submission time
user_type	String	Customer, vendor, internal
status	Enum	New, In-review, Escalated, Closed

Priority Entity

Field	Type	Description
priority_id	UUID	Unique ID
complaint_id	UUID	Foreign key
final_priority	Enum	Low / Medium / High / Critical
priority_score	Float	Weighted score
escalation_required	Boolean	Escalation flag

Feedback Entity

Field	Type	Description

Field	Type	Description
feedback_id	UUID	Unique ID
complaint_id	UUID	Foreign key
resolution_time	Integer	Time to close (hours)
sla_breached	Boolean	SLA violation
human_override	Boolean	Manual change
satisfaction_rating	Integer	Post-resolution rating

11.2 Schemas and migrations

- Relational schema implemented using PostgreSQL / MySQL
- Foreign key constraints ensure integrity
- Schema versioning used to support incremental updates
- Migrations tested on staging environment before deployment

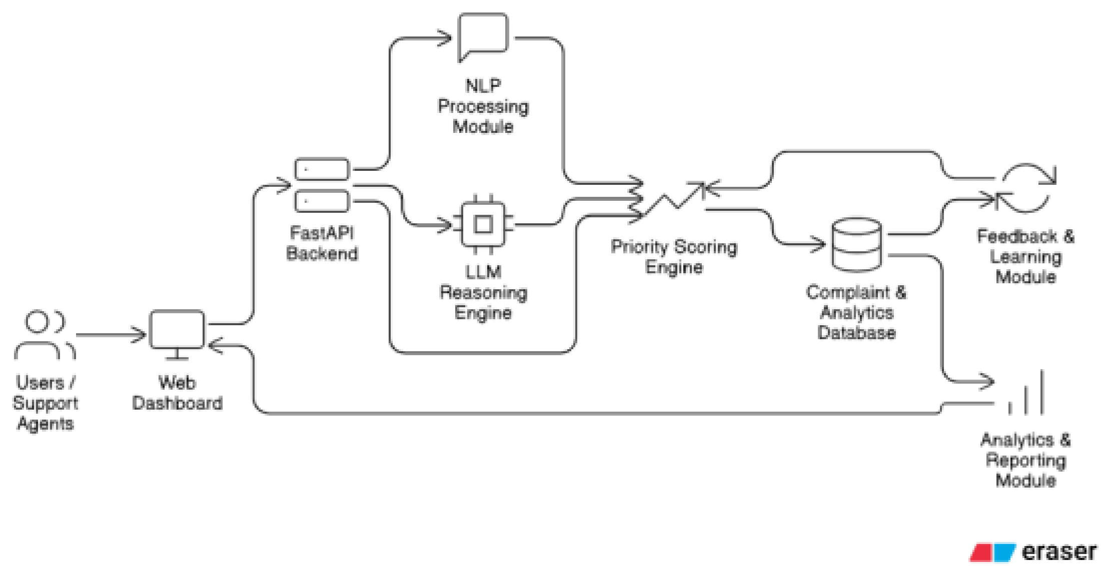
11.3 Privacy, retention, backup/DR

- Personally identifiable information (PII) is anonymized
- Complaint text stored with encryption at rest
- Retention policy: 12 months for academic usage
- Automated daily backups
- Recovery Time Objective (RTO): 4 hours
- Recovery Point Objective (RPO): 24 hours

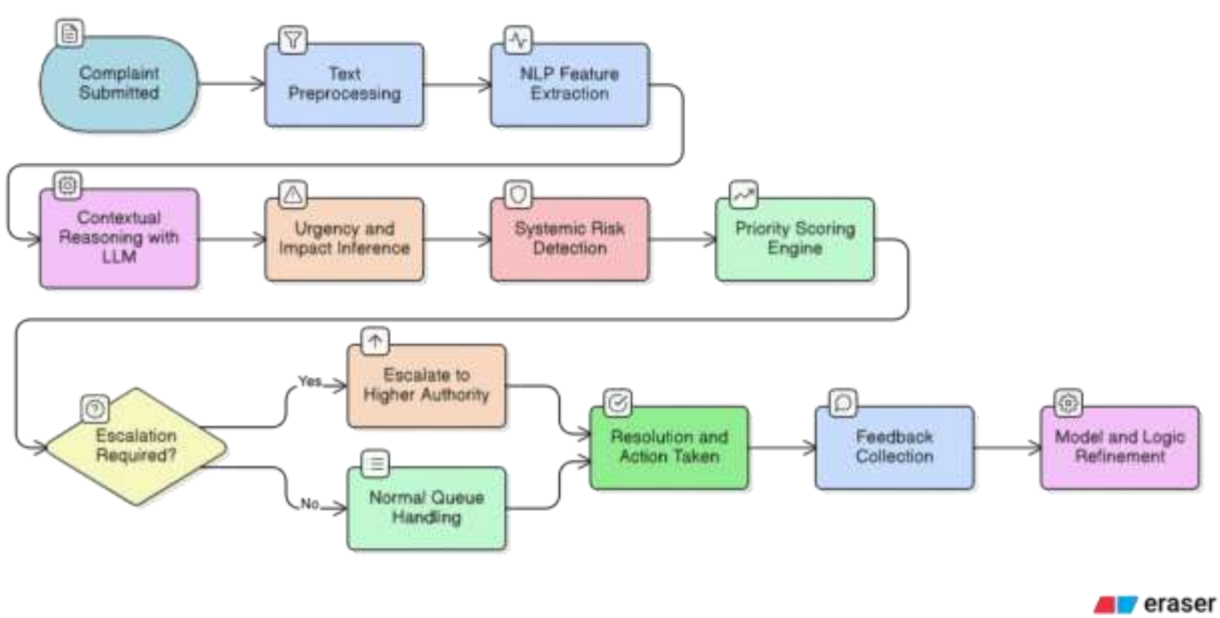
---

12. Technical Workflow Diagram

i. Work Architecture Diagram



ii. Data Flow Diagram



13. Quality: NFRs and Testing

13.1 Non-functional requirements

Metric	Description	Target
Availability	System uptime	≥ 99%
Latency	End-to-end analysis	≤ 2 seconds
Scalability	Complaints/day	≥ 10,000
Accuracy	Correct prioritization	≥ 85%
Explainability	Human readable	Mandatory

13.2 Test plan

Area	Type	Tool	Owner
NLP Module	Unit	PyTest	Ichcha
LLM Outputs	Validation	Custom checks	Gaurav
APIs	Integration	Postman	Balkrishna
Dashboard	UI Testing	Manual	Team

13.3 Environments

- Development: Local machines
- Staging: Integrated testing
- Production: Final demo environment

---

14. Security and Compliance

14.1 Threat model (STRIDE)

Asset	Threat	Impact	Mitigation
Complaint data	Data leak	High	Encryption
API endpoints	Injection	High	Input validation
LLM output	Hallucination	Medium	Rule checks

14.2 AuthN/AuthZ

- Role-based access control (RBAC)
- Roles: Agent, Manager, Admin
- Authorization enforced at API level

14.3 Audit and logging

- Complaint submission logs
- Analysis and escalation logs
- Feedback and overrides tracked
- Logs retained for 90 days

14.4 Compliance

- Academic data usage policy
- No third-party data sharing
- Ethical AI guidelines followed

---

15. Delivery and Operations

15.1 Release plan

Version	Features
v0.1	Complaint ingestion
v0.5	AI analysis & scoring
v1.0	Dashboard + feedback

15.2 CI/CD and rollback

- Continuous Integration: Test → Build
- Continuous Deployment: Versioned release
- Rollback supported via container images

15.3 Monitoring and alerting

Metric	Threshold	Action
Latency	> 2s	Investigate
Error rate	> 2%	Rollback
SLA breach	Any	Escalate

15.4 Runbooks

- **Latency spike:** Check DB → Reduce load → Rollback
- **Incorrect prioritization:** Review logs → Adjust weights → Reprocess

15.5 Communication plan

- Weekly mentor update
- Bi-weekly demos
- Bi-weekly demos

---

16. Risks and Mitigations

16.1 Risk heatmap

Risk	Probability	Impact	Mitigation
LLM hallucination	Medium	High	Validation rules
Cost overruns	Medium	Medium	Hybrid architecture
Bias in data	Low	High	Human review

---

17. Research and Evaluation

- Literature review: Existing complaint management systems and research on keyword-based ticketing, sentiment analysis, NLP-based classification, Explainable AI (XAI), and Large Language Models (LLMs) were studied to understand current limitations in complaint prioritization and escalation.
- Technology review: Transformer-based language models, contextual reasoning approaches, and hybrid AI architectures (ML + LLM) were analyzed for their suitability in inferring urgency, impact, and systemic risk from complaint text.
- Evaluation plan: System performance will be evaluated using KPIs such as prioritization accuracy, false escalation rate, SLA breach reduction,

latency, and explanation clarity. Results will be compared against rule-based and sentiment-based baselines.

- Feedback analysis: Post-resolution outcomes and human overrides will be reviewed to assess correctness of AI decisions and refine prioritization logic.
- Limitations: Evaluation is limited to text-based complaints, English language inputs, academic-scale datasets, and simulated operational environments.

---

18. Appendices

This section provides essential supporting information, definitions, and references relevant to the proposed Generative AI–based complaint prioritization system.

Glossary

Term	Description
NLP	Natural Language Processing – techniques used to analyze and understand complaint text.
LLM	Large Language Model – used for contextual reasoning and explanation generation.
SLA	Service Level Agreement – defines expected resolution time for complaints.
KPI	Key Performance Indicator – metrics used to evaluate system performance.
XAI	Explainable Artificial Intelligence – ensures transparency in AI decisions.
Systemic	A recurring or widespread issue affecting

Term	Description
Risk	multiple users or services.
Escalation	Promotion of a complaint to higher priority or authority based on risk assessment.

References

- Jurafsky, D., & Martin, J. H., *Speech and Language Processing*, Pearson.
- Vaswani et al., "Attention Is All You Need," NeurIPS.
- Ribeiro et al., "Why Should I Trust You?" – Explainable Artificial Intelligence.
- Official technical documentation of Large Language Model providers.
- Research literature on AI-based customer support and complaint analysis systems.